

KJMR2014

Unified Visual Perception Model and Its Application*

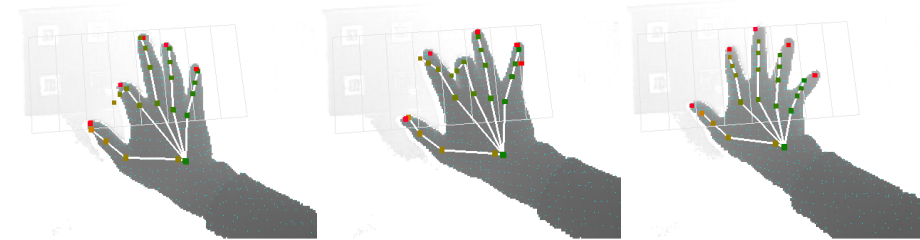
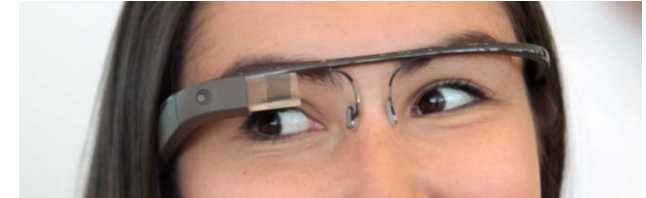
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UVR Lab. ICVL Lab. ICVL Lab. UVR Lab.

*Partly in Collaborating with Imperial College
London

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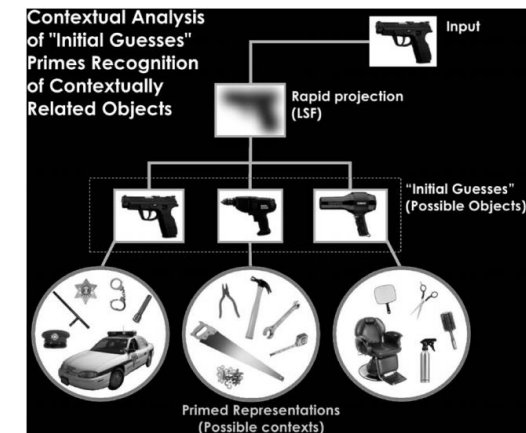
General Background

- According to the advances of
 - Optical see-through HMD
 - Glasses-type device
- There are rapidly increasing demands on AR technology,
 - which moves forward to “practical field application” considered in real situations.
- Challenges for ambiguous visual data
 - Partly occluded object identification
 - Self-occluded fingertip position estimation



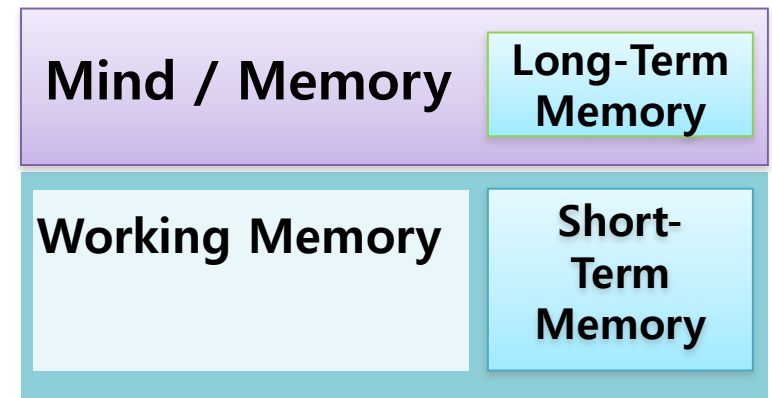
Motivation

- Why do algorithms make error, while human recognize well?
- Human visual perception is affected
 - Not only by low-level visual information processing
 - But also by experience and intention in a top-down manner
- Need a visual perception model
 - Integrating Bottom-up approach (low-level processing)
 - Top-down approach (predictive information processing) [1]
- Academic plausibility by adopting the results from the field of
 - Biological study (BU: low-level processing)
 - Cognitive science and brain research (TD: high-level processing)



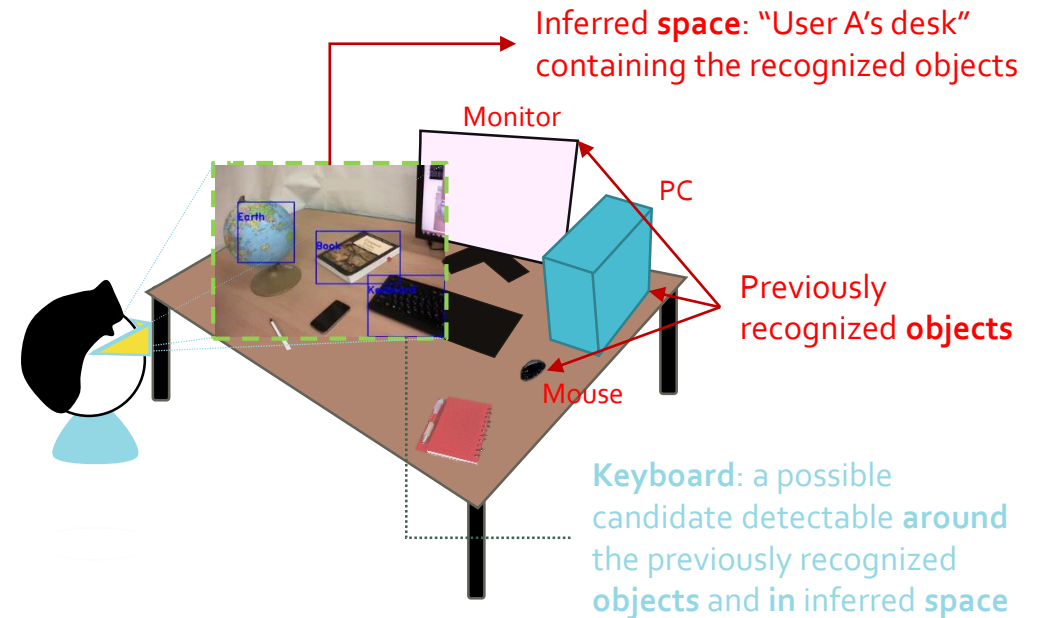
Summary of the Research

- Unified Visual Perception Model (UVPM)
 - Imitating the human visual perception process
 - Designed based on
 - Cognitive informatics
 - Brain research
 - Psychological science
- Consisting of (mutually complementary)
 - Working memory (low-level processing in a bottom-up manner)
 - Long/Short-term memory (high-level processing in a top-down manner)



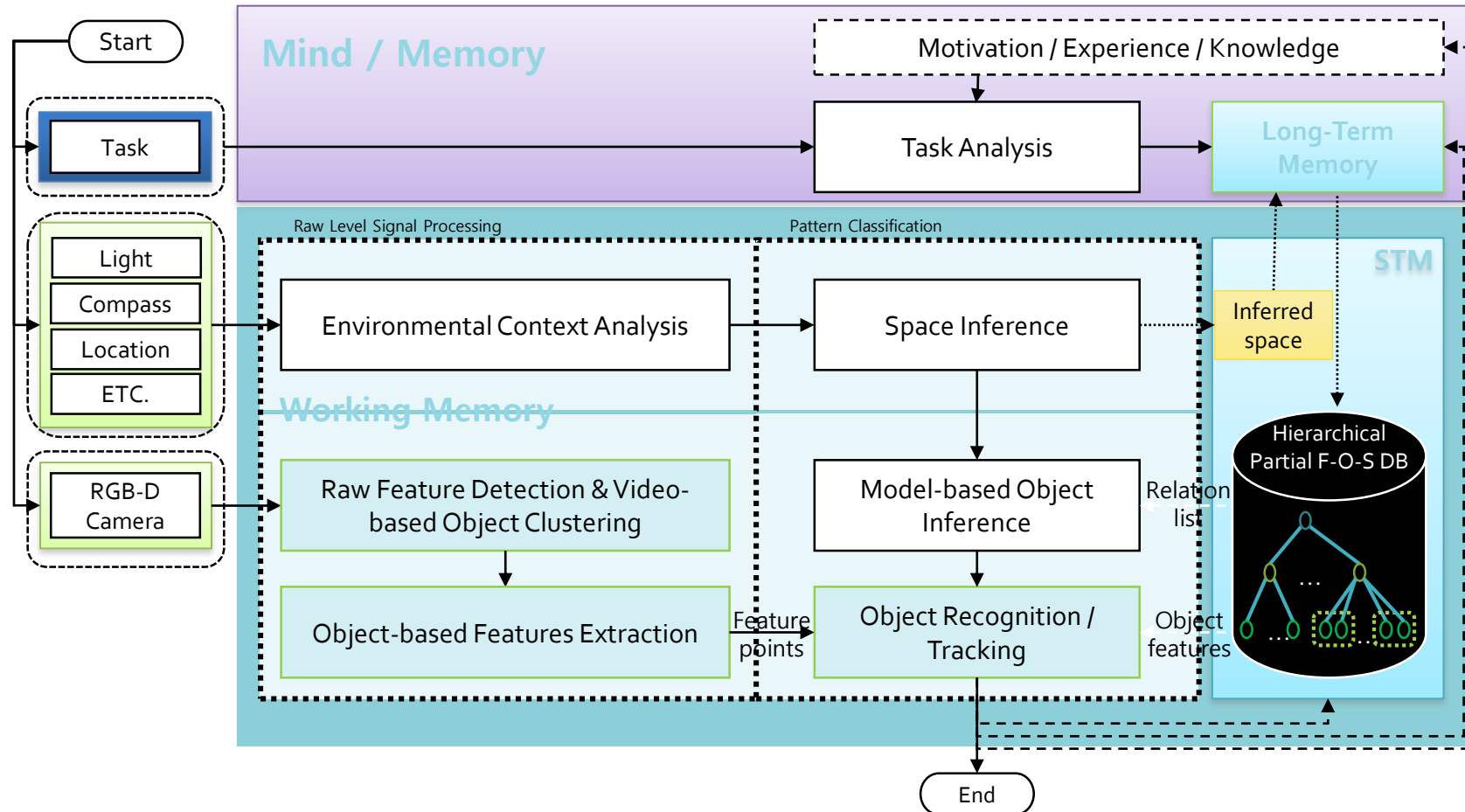
App. #1: Object Recognition

- Bottom-up approach
 - Need to consider how to configure the extracted features in a symbolic representation before focusing on making the features discriminable
- Top-down approach
 - Need to consider how to make and manage the semantic form as predictive information



UVPM for Object Recognition

• overview



UVPM for Object Recognition

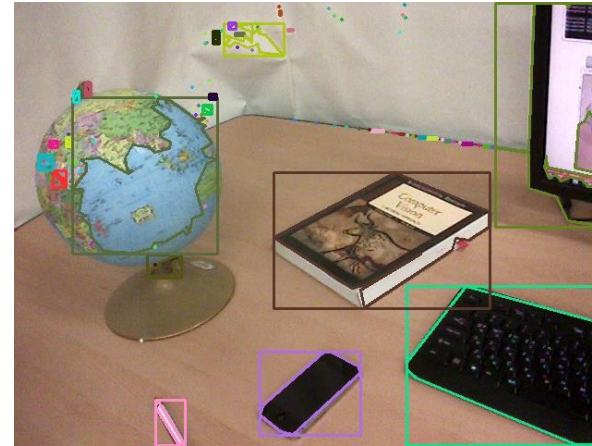
- *Environmental Context Analysis* and *Space Inference*

- Human can retrieve information related to the location where they are
- Assumption: the multiple sensors
 - that are built into current mobile devices are made available to use
- By analyzing the combination of the input data,
 - We can infer a candidate of space, which is one of the preliminarily defined many spaces
- the inferred space candidate is to be used an input for bringing a part of the hierarchical F-O-S tree from LTM to STM because searching range of the whole tree needs to be reduced for efficient search

UVPM for Object Recognition

- *Raw Feature Detection & Video-based Object Clustering*

- Extracting raw features
 - **Prototype:** saliency map detection (adopting histogram-based contrast method [1])
- Grouping the extracted raw features based on motion information
 - **Prototype:** Each salient region grouping by finding the closed contours based on [2]

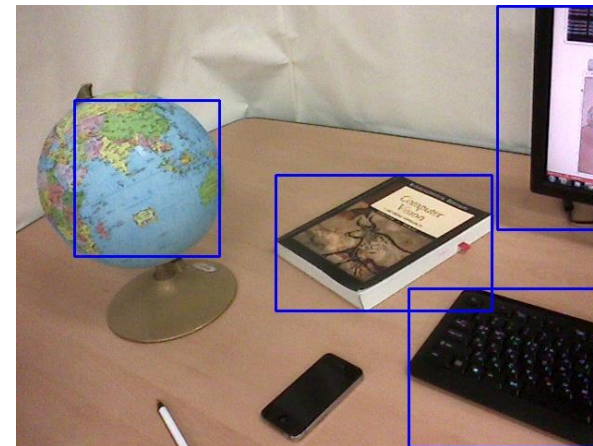
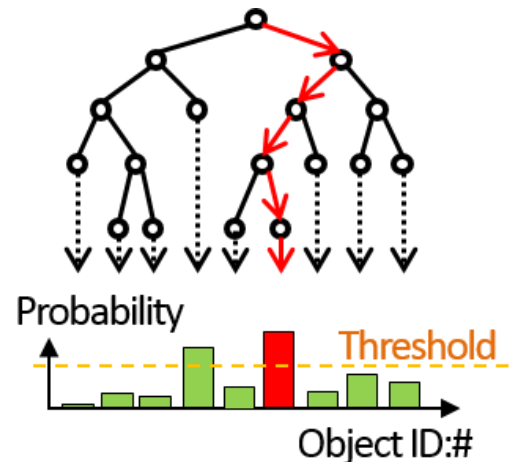


UVPM for Object Recognition

- *Object-based Features Extraction* and *Object Recognition/Tracking*
 - For the initial guesses of objects
 - Imitating the concept of coarse representation, as a symbolic form
 - By adopting bag-of-words (BoWs) method to represent an object
 - The reason we chose the BoWs
 - BoWs itself treats an image as a set of features (meaning its generalized representation)
- **Prototype:**
 - FAST corner detector
 - SURF descriptor
 - Randomized Trees as a codebook and a classifier

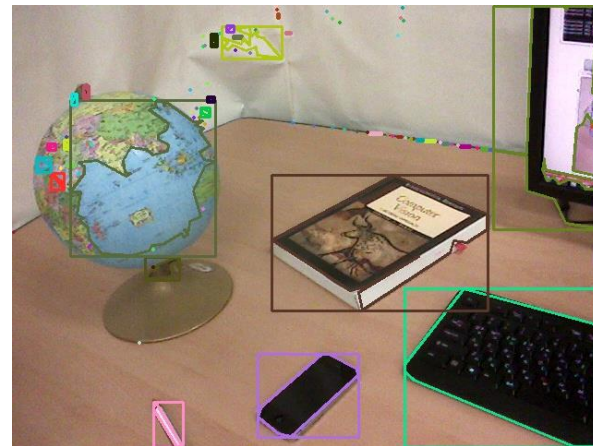
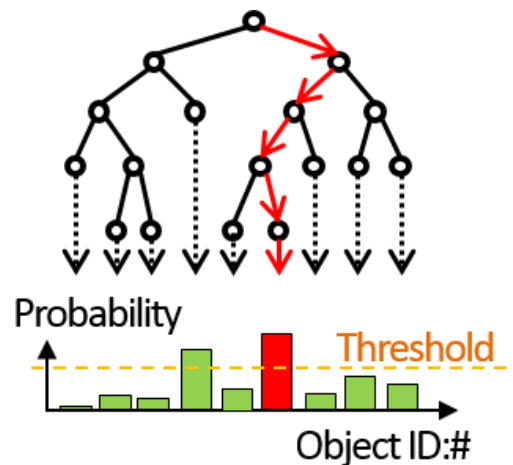
UVPM for Object Recognition

- **Object-based Features Extraction** and **Object Recognition/Tracking**
 - 1. BoWs for a ROI can be formed through the RF codebook
 - 2. BoWs are classified based on the pre-learned RF classifier
 - In this state, classifier determines the recognized object by selecting the object label that maximizes the probability among others



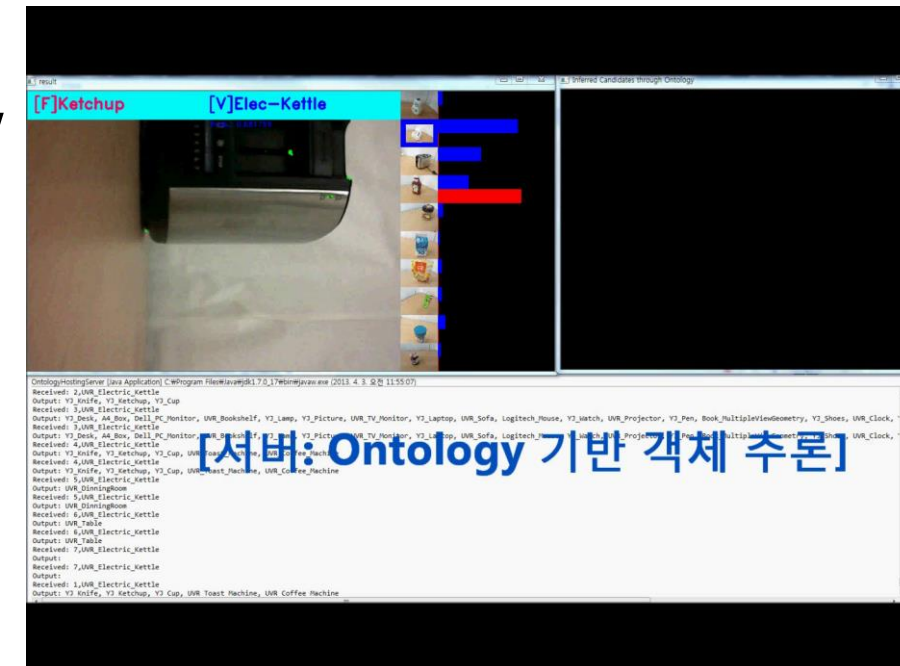
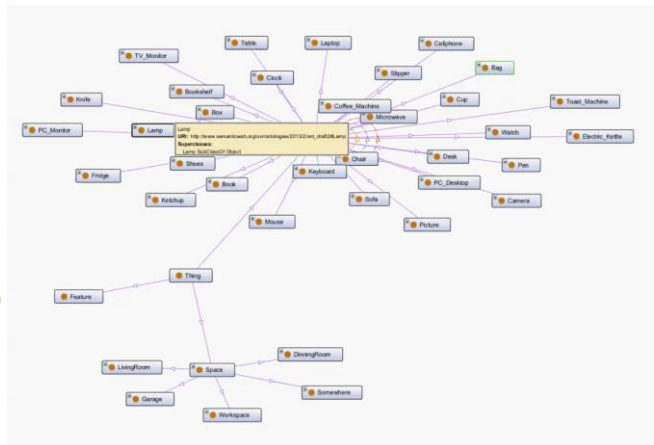
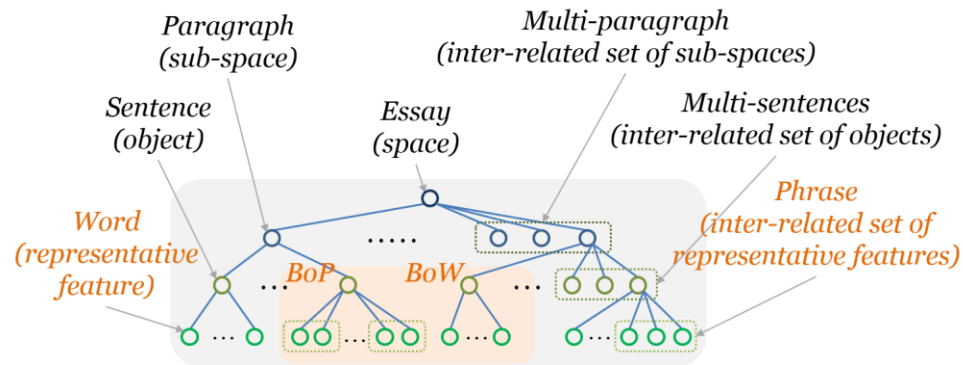
UVPM for Object Recognition

- **Object-based Features Extraction** and **Object Recognition/Tracking**
 - With all the probabilities for all the candidate ROIs,
 - Filtering ROIs
 - By measuring the size, the # of feature points extracted within a ROI
 - By selecting the regions where the probability is larger than the predetermined threshold



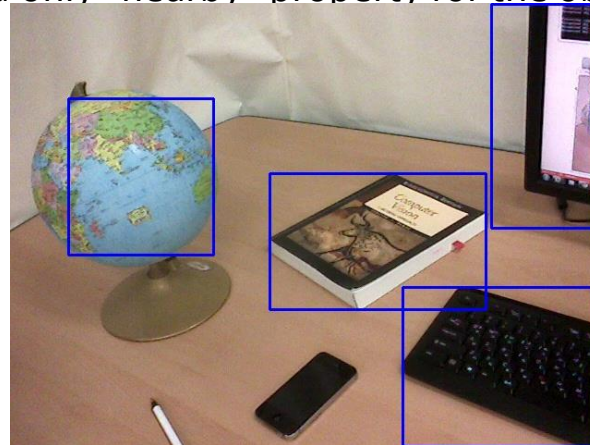
UVPM for Object Recognition

- **Long-term Memory** to **Short-term Memory**
 - Top-down approaches for the predictive information
 - By defining relationships among objects based on ontology
 - **Prototype:**
 - Ontology is modeled based on Java-based Protégé
 - Properties of each individuals: “nearby”, “on to”, “underneath”, “placed in”, “size”



UVPM for Object Recognition

- *Model-based Object Inference* and *Object Recognition/Tracking*
 - To confirm the effectiveness of LTM,
 - we defined somehow strong assumptions for the relationships objects.
 - To retrieve the information related to the initially recognized results from the knowledge-based framework,
 - **Prototype:**
 - TCP/IP socket communication between different programming language (C++)
 - Utilized only “nearby” property for the object recognition



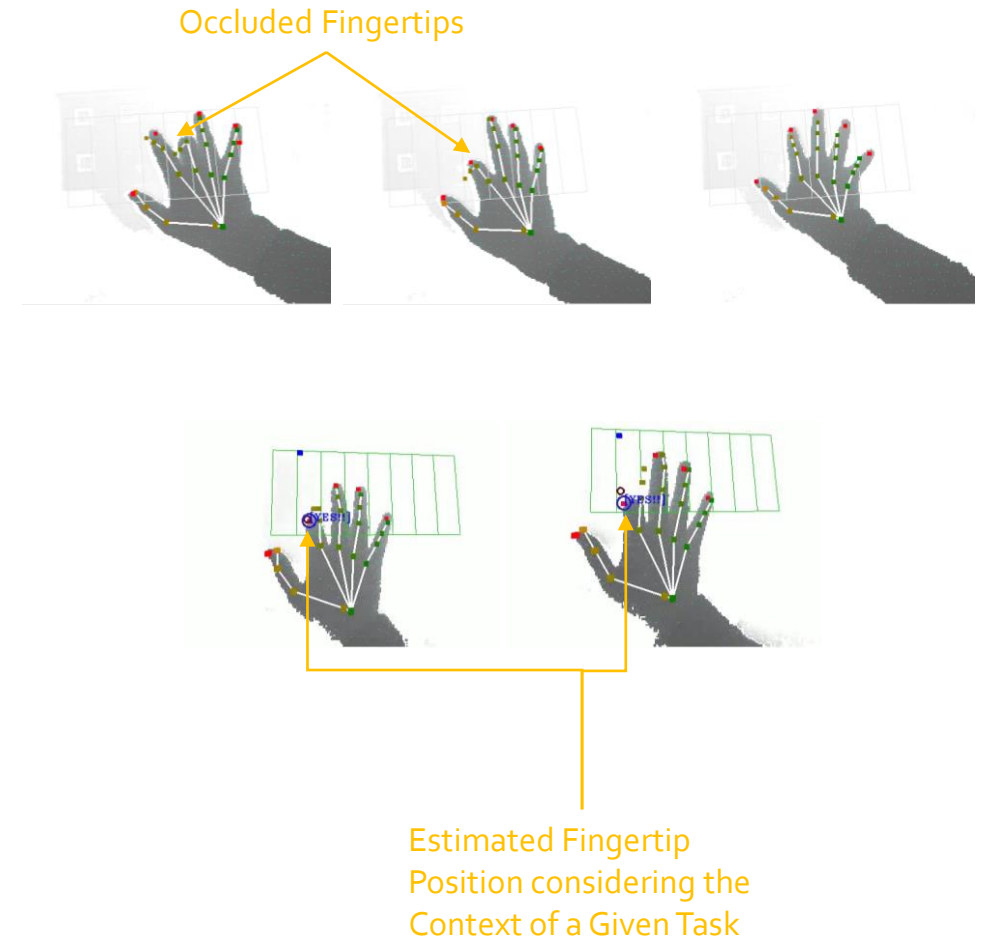
UVPM for Object Recognition

- *Task Analysis*

- This is a type of black box
 - that manages and processes the human mind or memory.
- Assumption: this certainly influences the development of LTM
 - based on motivation, experience, knowledge and so on.
- However, at this moment,
 - we have not yet found the apparent process of it.

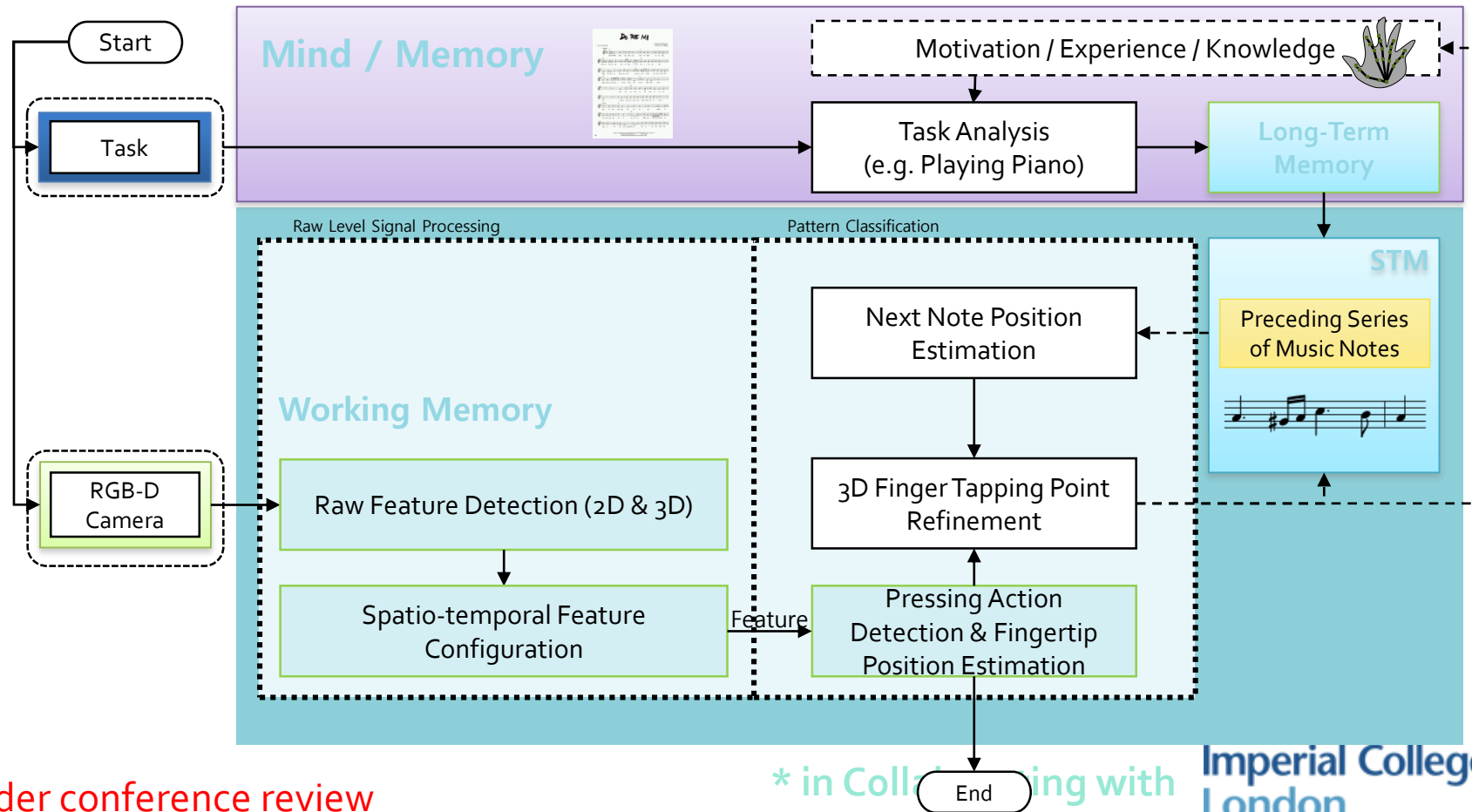
App. #2: 3D Finger Tapping Point Estimation (TPE)

- Bottom-up approach
 - Need to consider how to configure the extracted features as a spatio-temporal feature
 - 2D-3D spatio-temporal data fusion
- Top-down approach
 - Need to consider how to make and manage the semantic model as predictive information



UVPM for 3D Finger TPE

• overview



* The work is under conference review

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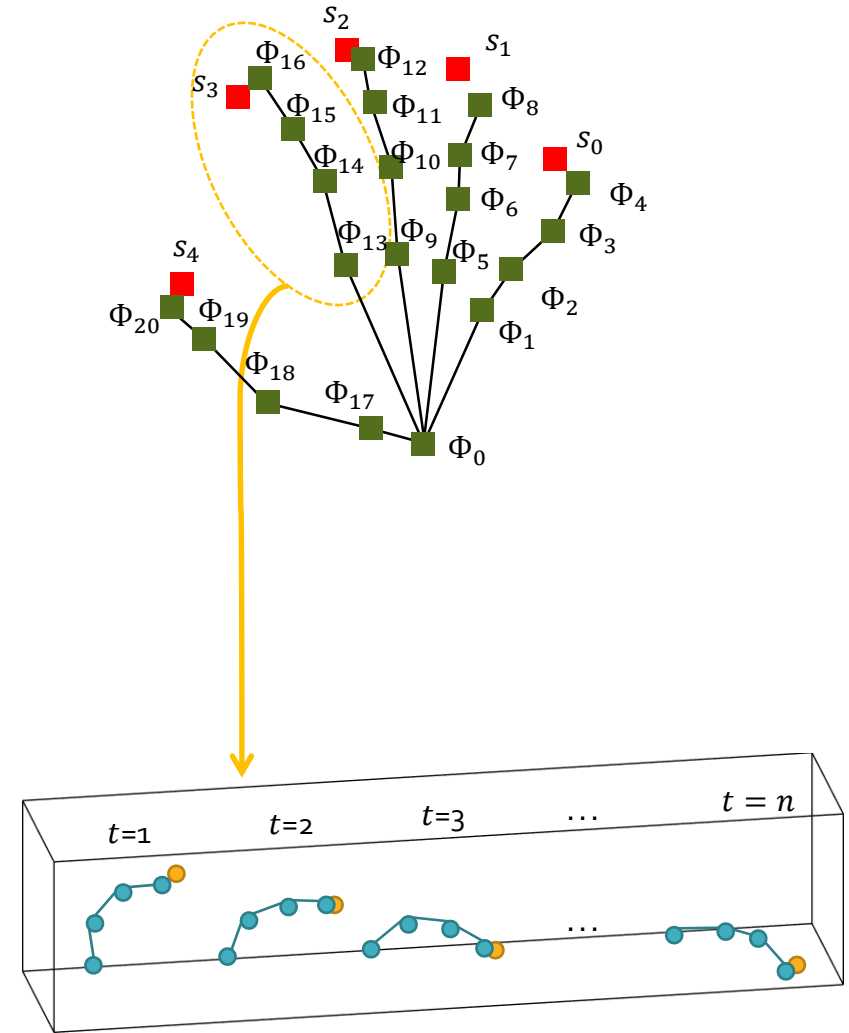
UVPM for 3D Finger TPE

- *Raw Feature Detection*

- Extracting raw features
 - 2D image-based fingertip detection
 - 3D Hand Pose Estimator [1]

- *Spatio-temporal Feature Configuration*

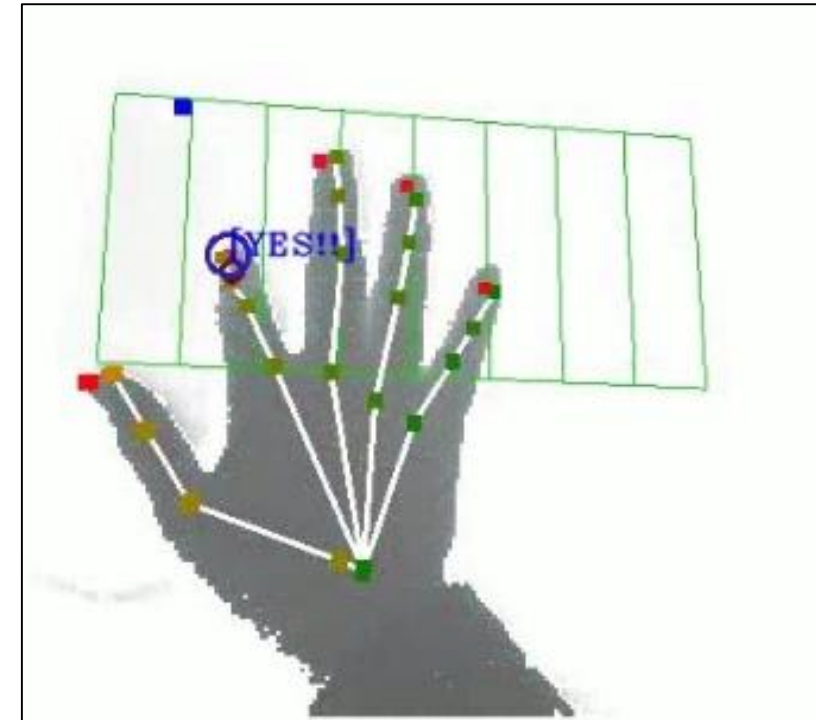
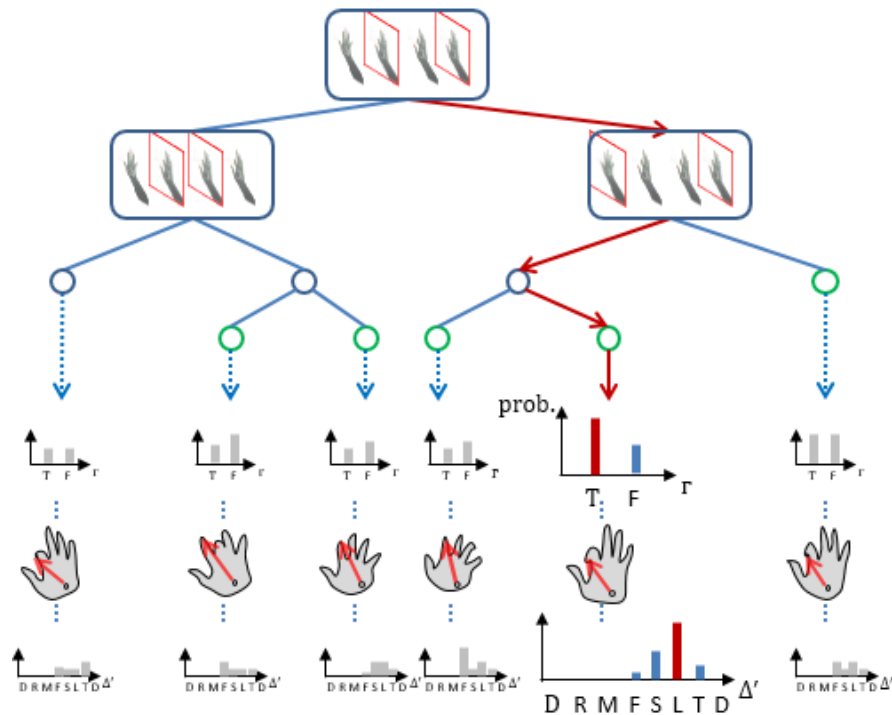
- Making a chunk containing frames captured within 500ms time period



* The work is under conference review

UVPM for 3D Finger TPE

- *Pressing Action Detection & Fingertip Position Estimation*



UVPM for 3D Finger TPE

- *Next Note Position Estimation & 3D TPE Refinement*

- Next Note Position Estimation
 - Adapting state transition model (e.g. HMM)
- 3D TPE Refinement
 - Multiplying the probabilities (State Transition Model + ST Forest)

Conclusion and Future Work

- We proposed “Unified Visual Perception Model”
 - Combining L/STM(TD) and WM(BU)
 - Reflecting the sequence of human visual processing
 - based on the theories of cognitive informatics, brain research, psychological science
- Prototyping each boxes shows development possibilities
 - In two different application scenarios
- We plan to
 - Verify the direction of the algorithm flows and the links between the boxes
 - Complete the proposed model by strengthening the theory base
 - from psychological and biological research areas
 - Experiment to prove its effectiveness
 - Conduct an application of context-aware wearable AR based on the proposed model